

AIQI ZHANG

 aiqizhang.org

 aiqi_zhang@berkeley.edu

 [linkedin.com/aiqi](https://www.linkedin.com/in/aiqi)

 github.com/aiqiz

Education

University of California, Berkeley

Aug 2026 - May 2027

Master of Science in Civil and Environmental Engineering

Berkeley, CA, USA

Major in Energy, Civil Infrastructure and Climate

University of Toronto

Sept 2022 - May 2026

Bachelor of Applied Science in Engineering Science with Honors

Toronto, ON, Canada

Major in Engineering Physics, Minor in Artificial Intelligence

NOIC academy

Sept 2021 - June 2022

Secondary School Diploma, Valedictorian Student of Year 2022

Markham, ON, Canada

The Affiliated High School of Peking University

Sept 2016 - June 2021

Junior Middle School Diploma

Beijing, China

Publications

Harlick, A. M., Wunch, D., Lee, C., Hopper, M., Banwait, P., Chen, J., Kou, M., Xue, Q., Espinas, C., and **Zhang, A.** *Adding Environmental Science into Advanced Physics Laboratories*. Physics in Canada. Manuscript under review.

Zhang, A., Restrepo-Coupe, N., Madsen-Colford, S., Ars, S., Weaver, D., Wu, D., Kandapath, M., Ma, W., Arain, A., and Wunch, D. *High Resolution Estimation of Urban Vegetation Carbon Uptake through Phenocam and Fluorescence-based Remote-Sensing Observational Framework*. Manuscript in preparation.

Restrepo-Coupe, N., Wunch, D., **Zhang, A.**, et al. *Phenology and the City: The Role of Vegetation in Monitoring Toronto's Atmospheric Emissions*. Manuscript in preparation.

Zhang, A., Xiang, A., van Rooyen, K., Spada, V., Wu, M., and Harlick, A. *Experimental Setups for the Domino Chain Effect*. Manuscript in preparation.

Research Experience

Research Trainee; Casual Research Assistant - University of Toronto May 2025 - June 2026

Supervisor: Dr. Debra Wunch

Toronto, ON, Canada

- Developed a satellite-camera fusion framework integrating urban phenocam imagery, satellite solar-induced fluorescence (SIF), land cover, and meteorological observations to characterize urban vegetation dynamics.
- Built computer vision pipelines to extract vegetation color indices from urban traffic-camera networks and quantify phenological transitions across heterogeneous landscapes.
- Implemented spatial downscaling approaches for satellite SIF retrievals to 10 m resolution and evaluated ecosystem productivity estimates through comparison with camera-based and flux-based approaches.
- Developing a first-author manuscript investigating urban vegetation phenology and productivity using multi-source remote sensing observations.

Research Intern - Nanyang Technological University

May 2024 - Sept 2024

Supervisor: Dr. Xiaodong Chen

Singapore

- Designed and optimized a large-scale battery charging-discharging cycle database framework for machine learning-based lifetime prediction.
- Developed physics-informed feature extraction methods integrating empirical battery degradation mechanisms with machine learning models to improve interpretability.
- Investigated the influence of end-of-life definitions on battery lifetime prediction accuracy through comparative computational analysis.

Teaching & Laboratory Experience

Graduate Student Instructor - University of California, Berkeley

Aug 2026 - Dec 2026

Physics 7B

Berkeley, CA, USA

- Graduate student instructor for physics department, with 50% (20 hours) employment.
- Teach undergraduate physics laboratories through experiment demo, holding office hours, grading, and preparation of teaching materials.

Lab Technician - University of Toronto

May 2025 - June 2026

Supervisor: Dr. Ania Harlick

Toronto, ON, Canada

- Supported undergraduate physics laboratories through experiment development, equipment troubleshooting, student instruction, and preparation of teaching materials.
- Adding Environmental Science Into Advanced Physics Laboratories
- Developed laboratory manuals and instructional resources to improve hands-on learning experiences in the Advanced Physics Laboratory course.
- Built and debugged experimental apparatuses through CAD design, electronics integration, software development, instrumentation, and iterative testing.

Conference Presentations

Zhang, A., Braverman, B., and Harlick, A.: *Computer Vision for Domino Chain Dynamics*, Women+ in Physics Canada, Kingston, ON, July 2025 (poster).

Projects

GPU-Accelerated Quantum Simulation

Jan 2026 - April 2026

Industry-Sponsored Engineering Design Project

- Collaborated with Nanoacademic Technologies to investigate GPU acceleration strategies for finite-element solvers used in quantum device modeling.
- Analyzed computational bottlenecks in Poisson and Schrödinger equation solvers and explored numerical optimization approaches to improve solver throughput and scalability.
- Designed computational strategies for integrating parallel computing approaches into existing simulation workflows across different hardware architectures.

Domino Chain Effect

Sept 2024 - June 2026

Advanced Physics Laboratory Project

Supervisor: Dr. Boris Braverman and Dr. Ania Harlick

- Developed a stereo three-dimensional reconstruction pipeline for high-speed domino dynamics using camera calibration, triangulation, and coordinate transformation techniques.
- Quantified six-degree-of-freedom chain reaction mechanics through computer vision-based color-marker tracking, revealing rotational dynamics beyond traditional one-dimensional models.
- Developed a physics-based framework to analyze wave propagation and energy transfer mechanisms in domino chain reactions.

Satellite Image Segmentation

Feb 2024 - April 2024

Deep Learning Course Final Project

- Developed a deep learning framework for semantic segmentation of satellite imagery using encoder-decoder architectures.
- Implemented data preprocessing, augmentation, and model optimization pipelines using PyTorch to improve classification performance.
- Explored computer vision approaches for geospatial and environmental data analysis.

PatchouLink

Jan 2024 - April 2024

Praxis III Engineering Design Project

- Designed and prototyped an IoT-based plant cultivation monitoring system integrating in-ground sensors, wireless communication, and real-time data visualization.
- Developed a full-stack monitoring platform using Arduino, Python-Django, and SQL databases for automated data processing and visualization.

OpenSOS Emergency Response AI Agent

Oct 2023

2023 NewHacks Hackathon — 1st Place, Smart Cities Category

- Developed an AI-powered emergency response agent capable of extracting critical information from callers and communicating relevant data to first responders.
- Implemented voice recognition and large language model-based analysis using Python libraries, GPT API integration, and prompt engineering techniques.

Betta-Trap

Jan 2023 - April 2023

Praxis II Engineering Design Project

- Designed and prototyped a storm-drain filtration system to prevent microplastic transportation into the Great Lakes while maintaining overflow protection.
- Led mechanical design and CAD modeling of a double-liner system with overflow mechanisms and maintenance accessibility features.

Scholarships and Honors

Samuel Beatty Physics Conference Travel Award, July 2025
Physics Summer Student Program Fellowship, May 2025

Technical Skills

Programming: Python, C/C++, MATLAB, SQL, R, HTML, JavaScript, CSS, Verilog, LaTeX

Machine Learning: PyTorch, TensorFlow, Computer Vision, Deep Learning, Data Processing

Remote Sensing: Satellite SIF, Phenocam Analysis, Vegetation Monitoring, Geospatial Data Processing

Hardware & Instrumentation: Arduino, Raspberry Pi, Scientific Instruments, Data Acquisition Systems

CAD & Prototyping: Fusion360, SolidWorks

Photography and Video Editing: Final Cut Pro, Adobe Photoshop, Adobe Lightroom, Adobe After Effects

Language: English, Mandarin

Relevant Coursework

Physics: Atmospheric Physics, Computational Physics, Advanced Classical Mechanics, Thermal Physics, Fluid Mechanics, Electromagnetism, Statistical Mechanics, Quantum Mechanics, Physics of the Earth, Advanced Physics Laboratory

Electrical and Computer Engineering: Machine Learning, Deep Learning, Algorithms, Data Structures, Electronics, Digital and Computer Systems, Electromagnetic Fields, Data Modeling Prediction and Control

Mathematics: Complex Analysis, Differential Equations, Vector Calculus, Probability and Statistics, Linear Algebra, Calculus I, Calculus II

Engineering Design: Praxis I&II&III Engineering Design Courses, Structural Engineering, Molecules and Materials, Biomedical Engineering, Engineering and Society, Economic Analysis and Decision Making

Design Team / Extracurricular

UTSM Urban Concept Aero-structural Team

Sept 2023 – May 2024

Team Member

University of Toronto

- Contributed to prototype development and CAD design of the outer shell for the first hydrogen fuel-cell vehicle prototype.

Natural Observation Team

Sept 2016 – Present

Team Member

Wings of Nature - Natural Observation Club

- Participated in ecological field observations and biodiversity surveys across multiple national conservation areas in China.
- Served as a tutor during insect observation field activities at Gaoligong National Nature Reserve in July 2023.